

Similarly, the second professor would have answered No if he wanted no coffee. Given his answer, his non-knowledge about the 3rd prof, we imply he wants coffee.

The third professor realizes the 1st two preferences. Since the reply is No, he wants no coffee.

So the hostess gives coffee only to the 1st and 2nd professor

22) When planning a party you want to know whom to invite. Among the people you would like to invite are three touchy friends. You know that if Jasmine attends, she will become unhappy if Samir is there, Samir will attend only if Kanti will be there, and Kanti will not attend unless Jasmine also does. Which combinations of these three friends can you invite so as not to make someone unhappy?

Soln: p : Jasmine attends the party q : Samir attends the party
 r : Kanti attends the party

① $q \rightarrow \neg p$ ② $q \rightarrow r$ ③ $r \rightarrow p$ ~~are True~~ are True

Let q be True. $\therefore$ By ②, and defn. of conditional, r is True

By ③, and defn of conditional, p is True.

By ①, and defn of conditional, $\neg p$ is True. This is a contradiction.

$\therefore q$ must be False. ~~Jasmine and Samir can attend together~~

Jasmine, Kanti can attend together. Samir, if invited is alone

Exercises 23-27 relate to inhabitants of the island of Knights and Knaves created by Smullyan, where Knights always tell the truth and Knaves always lie. You encounter two people, A and B. Determine, if possible, what A and B are if they address you in the ways described. If you cannot determine what these two people are, can you draw any conclusions?

23) A says "At least one of us is a knave" and B says nothing.